MineMind

JV Inkai: acid delivery timing and wellfield commissioning

Which new block groups start later when acid arrives late?

A source-informed, representative digital twin on OpenEgiz.
18 twins. One adjustable scenario. Complete reset.

Synthetic block groups and schedules. Assumed commissioning response.
No site calibration, optimizer or actual operating forecast.

Connected platform and mine flows

Model publisher -> MQTT -> Eclipse Ditto -> spatial interface

The UI reads all operating states and projections back from Ditto.
Missing twins or mixed revisions produce an error, not substitute model data.

Commissioning acid

Separate area stocks -> whole-dose reservation -> acidification -> commissioning

Uranium-bearing solution and product

Block extraction -> area IX -> recovered-U queue -> shared MPP drying

Existing/sustaining acid supply is protected outside this model. Satellite transfer is aggregated.

Evidence and explicit assumptions

Public facts

MPP / Sat-1 / Sat-2 IX: 2.7 / 6.3 / 4.5 Mlb U3O8/year.
Drying: 8.3 Mlb/year. Plant recovery: approximately 98%.

Synthetic exercise and assumptions

Nine group identities, positions, stocks, rates and schedules.
Whole commissioning doses: 100-160 t per new group.
Seven-day acidification, seven-day ramp, fixed 100 mg U/L.
Annual capacities converted to uniform daily model limits.

The operator reported new-block acidification delays despite 106% nine-month acid fulfilment. This supports the timing mechanism, not these coefficients or a 28-day reconstruction.

Sources: Cameco 2024 Inkai report, printed pp.80-82; Kazatomprom 3Q24 update; PARAMETERS.md

Accountable inventories and constraints

Acid stock = initial stock + deliveries - reserved doses

Extracted U = dried product U + queue U + unrecovered U

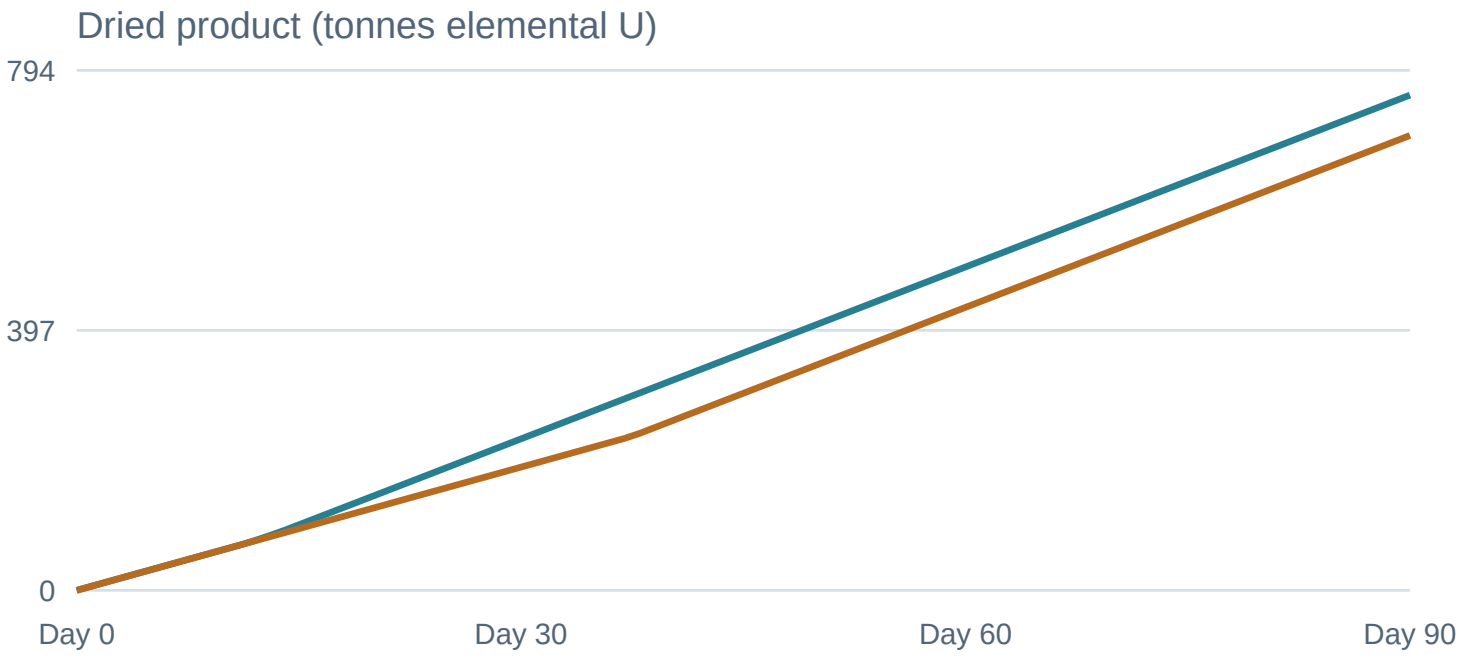
Initial group U = remaining U + cumulative extraction

Area IX limits proportionally throttle extraction.
Recovered uranium remains in the drying queue when packaging binds.
The 2% unrecovered stream is recorded; uranium is not destroyed.

Conservation tests establish internal consistency, not real-world operational validation.

Simulated delivery-delay comparison

First arrivals: day 3 -> day 31 (+28 days). Second arrivals stay day 60.



Baseline

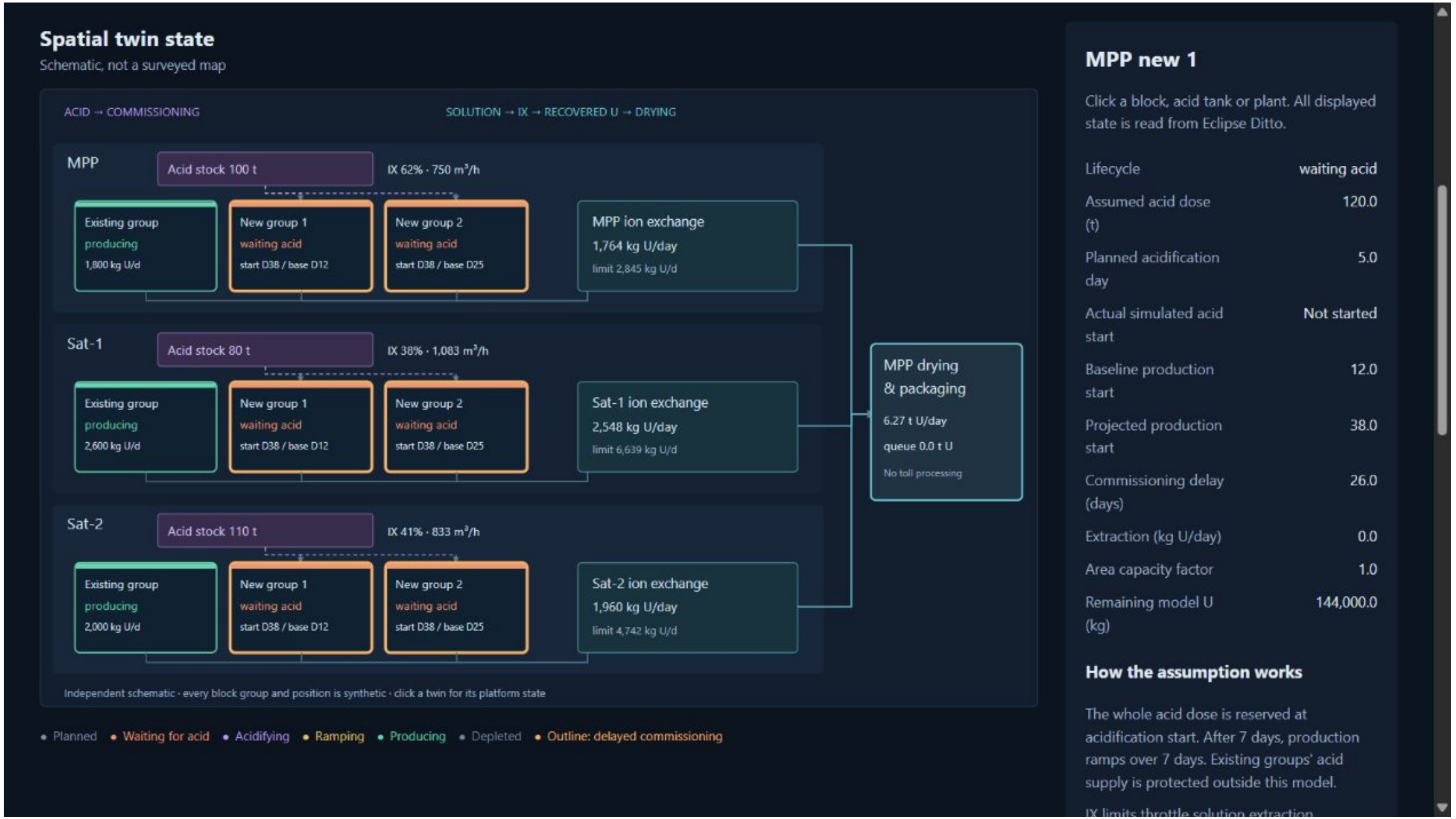
28-day delay

Day 30: 231.30 vs 188.16 t U.
Day 90 difference:
-61.66 t U.

New B1 starts: D12 -> D38.
Same total acid by day 90.

A model outcome, not an Inkai loss estimate. Different assumptions or bottlenecks can change the difference.

Generated from model.py/config.json by build_presentation.py; no observed block data or calibration


Spatial twin state
Schematic, not a surveyed map
ACID → COMMISSIONING
SOLUTION → IX → RECOVERED U → DRYING
MPP
Acid stock 100 t
IX 62% · 750 m³/h
Existing group
producing
1,800 kg U/d
New group 1
waiting acid
start D38 / base D12
New group 2
waiting acid
start D38 / base D25
MPP ion exchange
1,764 kg U/day
limit 2,845 kg U/d
Sat-1
Acid stock 80 t
IX 38% · 1,083 m³/h
Existing group
producing
2,600 kg U/d
New group 1
waiting acid
start D38 / base D12
New group 2
waiting acid
start D38 / base D25
Sat-1 ion exchange
2,548 kg U/day
limit 6,639 kg U/d
Sat-2
Acid stock 110 t
IX 41% · 833 m³/h
Existing group
producing
2,000 kg U/d
New group 1
waiting acid
start D38 / base D12
New group 2
waiting acid
start D38 / base D25
Sat-2 ion exchange
1,960 kg U/day
limit 4,742 kg U/d
MPP drying
& packaging
6.27 t U/day
queue 0.0 t U
No toll processing
Independent schematic · every block group and position is synthetic · click a twin for its platform state
Planned
Waiting for acid
Acidifying
Ramping
Producing
Depleted
Outline: delayed commissioning
MPP new 1
Click a block, acid tank or plant. All displayed state is read from Eclipse Ditto.
Lifecycle waiting acid
Assumed acid dose (t) 120.0
Planned acidification day 5.0
Actual simulated acid start Not started
Baseline production start 12.0
Projected production start 38.0
Commissioning delay (days) 26.0
Extraction (kg U/day) 0.0
Area capacity factor 1.0
Remaining model U (kg) 144,000.0
How the assumption works
The whole acid dose is reserved at acidification start. After 7 days, production ramps over 7 days. Existing groups' acid supply is protected outside this model.
IX limits throttle solution extraction

Verification and reproduction

21 unit tests (Windows + Ubuntu)	PASSED
Live MQTT / Ditto / scenario / reset	PASSED
Original platform end-to-end smoke	PASSED
Browser spatial controls	PASSED
Fresh credentials and data-volume startup	PASSED

Start: sudo apt-get update && sudo apt-get install -y make
make inkai-test | make inkai | make inkai-smoke

Open localhost:8088. Reset -> view day 30 -> apply delay 28.
Inspect day 38, then day 90. Reset baseline to undo completely.

Local test: Ubuntu 24.04 WSL2 / Docker 29.1.3 / Compose 2.40.3 / about 7 GiB RAM.
Fresh-volume testing reuses downloaded images. Exact 16-GiB KVM judge reproduction remains unverified.
Detailed evidence, sources, assumptions and known limitations are committed alongside the code.